PARTIAL DIFFERENTIAL EQUATIONS

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Abstract. This paper is an overview of the Laplace transform and its applications to partial differential equations. We will present a general overview of the Laplace transform, a proof of the inversion formula, and examples to illustrate the usefulness of this technique in solving PDE's.

INTRODUCTION

The Laplace transform can be helpful in solving ordinary and partial differential equations because it can replace an ODE with an algebraic equation or replace a PDE with an ODE. Another reason that the Laplace transform is useful is that it can help deal with the boundary conditions of a PDE on an infinite domain. In this introductory section, we discuss definitions, theorems, and properties of the Laplace transform. Note that the proofs in this section are omitted, however if the reader is so inclined, the details are given in many standard texts on complex analysis and integral transforms [1].

1. Definition of the Laplace transform

To begin, note that the term *transformation* is often used in mathematics to mean a mapping $f : X \to Y$ where X and Y are sets. In general, X and Y need not refer to the same set. An example that illustrates this is the derivative, $D : F(x) \to F(x)$, acting on the set of differentiable functions over $\mathbb{R}$. Another transformation of functions that commonly arises is integration. This is denoted by:

$$I\{F(x)\} = \int_0^t F(x)dx$$

Next recall that a transformation T is called linear if

$$T(c_1\vec{v}_1 + c_2\vec{v}_2) = c_1T(\vec{v}_1) + c_2T(\vec{v}_2)$$

Notice that the derivative and the integral are two such transformations, when restricted to differentiable and integrable functions respectively. Next we define a general integral transform.

Definition 1.1. Let f be an integrable function of variable t over the interval (a, b) where a and b are in the extended real numbers. If $K(t, s)$ is a function of the variable t and parameter s, then the general *integral transform* of f with respect to the kernel $K(t, s)$ is represented by:

$$\int_a^b K(t, s)f(t)dt = F(s)$$

[1] See Donald W. Trim's text, *Complex analysis and Its Applications*

We shall see that with the careful choice of kernel, an ODE can be transformed into an algebraic expression, and a PDE can be transformed into and ODE. Before continuing, consider the following theorem:

Theorem 1.2. *Let T be an integral transform with kernel $K(t,s)$. That is*

$$T\{f(t)\} = \int_a^b K(t,s)f(t)dt.$$

Then T is a linear operator on the functions for which it is defined.

Choosing a suitable kernel can change a difficult problem into a more amenable one. Recall that the exponential function e^{-zt} is common in the solution of linear differential equations, where $z = s + i\omega$ is a complex parameter. When $\omega = 0$ this expression is simply

$$e^{-st},$$

and when $s = 0$ this becomes

$$e^{-\omega t} = \cos(\omega t) + i\sin(\omega t).$$

In the language of integral transforms, the Laplace transform takes the real part of this expression, namely e^{-st}, as its kernel, and the Fourier transform takes $e^{-i\omega t}$ as its kernel. Informally, engineers and scientists use the Laplace transform in studying the long term stability of a system, whereas the Fourier transform is helpful in studying a system that can be broken into a spectrum of *normal modes*.

This paper will be primarily concerned with the Laplace transform and its applications to partial differential equations. Therefore, without further discussion, the Laplace transform is given by:

Definition 1.3. Let f be a function of t. The Laplace transform of f is defined to be

$$F(s) = \int_0^\infty e^{-st}f(t)dt \tag{1.1}$$

provided the improper integral converges. When the Laplace transform exists, it is denoted by $\mathcal{L}\{f(t)\}$

The last line of our definition hints at the fact that this integral might not always converge. It is natural to ask if there exists a set of conditions that would guarantee convergence. The following definition is motivated by such a question.

Definition 1.4. A function, f, is said to be of *exponential order* α if there exist constants T and $M > 0$ in $\mathbb{R}$ such that $|f(t)| < Me^{\alpha t}$ for all $t > T$. This is denoted as $f(t) = O(e^{\alpha t})$.

Using the terminology above, we can now provide sufficient condition under which the Laplace transform is defined.

Theorem 1.5. *Let f be a function of t that is of exponential order α and is piecewise continuous on $0 \le t \le T$ for all $T \in \mathbb{R}$. Then $F(s) = \mathcal{L}\{f\}$ exists for all $s \ge \alpha$.*

Remark 1.6. Notice that when $\mathcal{L}\{f\}$ exists, it is unique because it is given by a convergent integral. However, given $F(s)$, there can be countably many functions f_i such that $\mathcal{L}\{f_i(t)\} = F(s)$. This follows from the fact that if two functions, f, g, are Lebesgue integrable over a set $A \subseteq \mathbb{R}$ and $f = g$ *almost everywhere*, then $\int_A f\, d\lambda = \int_A g\, d\lambda$.

2. Properties of the Laplace transform

In this section, we discuss some of the useful properties of the Laplace transform and apply them in example 2.3.

Theorem 2.1. *Let f be a continuous function of t with a piecewise-continuous first derivative on every finite interval $0 \leq t \leq T$ where $T \in \mathbb{R}$. If $f = O(e^{\alpha t})$, then $\mathcal{L}\{f'(t)\}$ exists for all $s > \alpha$, and*

$$\mathcal{L}\{f'(t)\} = sF(s) - f(0) \tag{2.1}$$

This theorem immediately leads to a result about second derivatives.

Corollary 2.2. *Let f and f' be a continuous function of t with a piecewise-continuous second derivative on every finite interval $0 \leq t \leq T$ where $T \in \mathbb{R}$. If f and f' are $O(e^{\alpha t})$, then $\mathcal{L}\{f''(t)\}$ exists for all $s > \alpha$, and*

$$\mathcal{L}\{f''(t)\} = s^2F(s) - sf(0) - f'(0) \tag{2.2}$$

To see how this can be useful in solving an initial value ODE, consider the following example:

Example 2.3. Consider the initial value problem:

$$u_{tt} + \omega^2 u = 0, \qquad u(0) = 0, \qquad u_t(0) = \omega$$

Solution: If we apply the corollary above, we can apply the Laplace transform to both sides of the ODE giving:

$$\mathcal{L}\{u_{tt} + \omega^2 u\} = \mathcal{L}\{0\}$$

Since $\mathcal{L}$ is linear, the above expression can be simplified to:

$$\mathcal{L}\{u_{tt}\} + \omega^2\mathcal{L}\{u\} = 0$$

by Corollary 2.2, we have

$$s^2U(s) - su(0) - u_t(0) + \omega^2U(s) = 0$$

Substituting initial conditions give the following:

$$s^2U(s) - \omega + \omega^2U(s) = 0$$

$$\left(s^2 + \omega^2\right)U(s) = \omega$$

Solving for $U(s)$ we get:

$$U(s) = \frac{\omega}{s^2 + \omega^2}$$

The solution, $u(t)$, of the system, is found by inverting the Laplace transform $U(s)$. According to Table 1, we have

$$\mathcal{L}^{-1}\{U(s)\} = \sin(\omega t)$$

This is the solution that one would obtain using elementary solution methods.

The example above shows that the Laplace transform changed our problem into basic algebra. However, it would be misleading to conclude that we have simplified our study of ODE's to the study of algebra. The main difficulty in applying the Laplace transform is in fact finding an inversion formula. Scientists and engineers often use a table of common transforms to invert a particular solution. The table below gives a list of such transforms:

Table 1

	$f(t)$	$F(s)$	$a \ \ (s > a)$
1	1	$\frac{1}{s}$	0
2	e^{at}	$\frac{1}{s-a}$	a
3	$t^n (n = 1, 2, ...)$	$\frac{n!}{s^{n+1}}$	0
4	$t^n e^{at} (n = 1, 2, ...)$	$\frac{n!}{(s-a)^{n+1}}$	a
5	$\sin kt$	$\frac{k}{s^2+k^2}$	0
6	$\cos kt$	$\frac{s}{s^2+k^2}$	0
7	$\sinh kt$	$\frac{k}{s^2-k^2}$	$\lvert k \rvert$
8	$\cosh kt$	$\frac{s}{s^2-k^2}$	$\lvert k \rvert$
9	$e^{-at} \sin kt$	$\frac{k}{(s+a)^2+k^2}$	$-a$
10	$e^{-at} \cos kt$	$\frac{s+a}{(s+a)^2+k^2}$	$-a$
11	$\sqrt{t}$	$\frac{\sqrt{\pi}}{2\sqrt{s^3}}$	0
12	$\frac{1}{\sqrt{t}}$	$\frac{\sqrt{\pi}}{s}$	0

Table 2. Table taken from [Churchill, p.17]

Using the table above, it is possible to find the inversion of many functions. The method of partial fractions and convolutions are particularly useful when combined with the information in table 1. In applications, situations arise in which $F(s)$ cannot be inverted using the methods mentioned above. Therefore, it is of great importance to have a general inversion formula for the Laplace transform. We will find that such a formula will involve integration in the complex plane. The next section is devoted to finding such a formula.

THE INVERSION FORMULA

As stated in the previous section, finding the inverse of the Laplace transform is the difficult step in using this technique for solving differential equations. For elementary problems, the use of Table 1 is often enough. However there arises the need for a general inversion technique when dealing with more complicated expressions. The following presentation gives a proof of the inversion formula. In this formulation, it is presumed that the reader has some familiarity with complex analysis, and when proofs are omitted, we refer the reader to Churchill [5].

3. Results From Complex Analysis

We begin with results from complex variables.

Theorem 3.1. *Let f be function of complex variable z, let C be a piecewise smooth curve in the complex plane, let M be the maximum of $|f(z)|$ on C, and let L be the length of C, then:*

$$\left| \int_C f(z)dz \right| \leq \int_C |f(z)||dz| \leq ML$$

This result is easy to show and will be used throughout this paper.

Theorem 3.2 (Cauchy Integral Formula)**.** *When f is analytic inside and on a simple, closed, piecewise-smooth curve C, its value at any point z interior to C is given by the contour integral*

$$f(z) = \frac{1}{2\pi i} \oint_C \frac{f(\zeta)}{\zeta - z} d\zeta. \tag{3.1}$$

Proof: *Refer to* [5]

The following theorem is a result of the Cauchy integral formula.

Theorem 3.3. *Let f be a function of complex variable s. If f is analytic in the half-plane $x > \delta$ where $\delta > 0$, and in this half-plane, there exist constants M, R, and $k > 1$ such that*

$$|f(s)s^k| < M \quad \textit{for} \quad |s| > R$$

And if z is a point such that $\gamma < Re(z)$ where $\gamma > \delta$, then

$$f(z) = -\frac{1}{2\pi i} \lim_{\beta \to \infty} \int_{\gamma - \beta i}^{\gamma + \beta i} \frac{f(\zeta)}{\zeta - z} d\zeta \tag{3.2}$$

Proof: This result is proved through the application of Theorem 3.1 along with the Cauchy integral formula and is very straight forward. For more details please *refer to* [5]

This result will play a key role in the development of an inverse formula.

4. The Inverse Laplace Transform

Theorem 4.1. *If f is a function of exponential order α, that is $O(e^{\alpha t})$, and f is piecewise continuous on every finite interval $0 \leq t \leq T$, then the Laplace transform $\mathcal{L}\{f(t)\} = F(s)$ of f is an analytic function of s in the half-plane $x > \alpha$.*

Proof: *Refer to* [5]

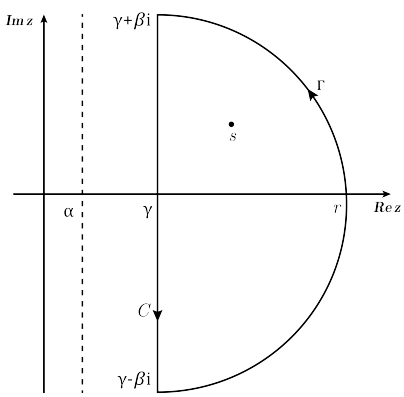
Theorem 4.2. *Let f be piecewise continuous on every finite interval $0 \leq t \leq T$, with $f(t) = O(e^{\alpha t})$. Suppose also that in some half-plane $x > \delta > 0$ there exists constants M, R, and $k > 0$ such that $|F(s)s^k| < M$ for $|s| > R$. If the inversion integral of F converges, then*

$$f(t) = \mathcal{L}^{-1}\{F(s)\} = \frac{1}{2\pi i} \lim_{\beta \to \infty} \int_{\gamma - \beta i}^{\gamma + \beta i} e^{st} F(s) ds$$

Proof. Since f is of exponential order α and f is piecewise continuous on every finite interval $0 \leq t \leq T$, we may apply Theorem 4.1 to conclude that F is analytic in the half-plane $x > \alpha$. Now using the fact that in some half-plane $x > \delta > 0$ there exists constants M, R, and $k > 0$ such that $|F(s)s^k| < M$ for $|s| > R$, we may apply Theorem 3.2 to obtain:

$$F(s) = -\frac{1}{2\pi i} \lim_{\beta \to \infty} \int_{\gamma - \beta i}^{\gamma + \beta i} \frac{F(\zeta)}{\zeta - s} d\zeta \tag{4.1}$$

This integral is in fact the application of the Cauchy integral formula over the following contour.



This is easily seen since

$$\lim_{\beta \to \infty} \int_C \frac{F(\zeta)}{\zeta - s} ds = -\lim_{\beta \to \infty} \int_{\gamma - \beta i}^{\gamma + \beta i} \frac{F(\zeta)}{\zeta - s} d\zeta \qquad \text{and} \qquad \lim_{\beta \to \infty} \int_\Gamma \frac{F(\zeta)}{\zeta - s} ds = 0$$

If we take the inverse transforms of both sides of equation (4.1), we obtain:

$$f(t) = -\frac{1}{2\pi i} \lim_{\beta \to \infty} \int_{\gamma - \beta i}^{\gamma + \beta i} -F(\zeta) \mathcal{L}^{-1}\left\{\frac{1}{s - \zeta}\right\} d\zeta \tag{4.2}$$

Now by line 2 of Table 1, we see that

$$\mathcal{L}^{-1}\left\{\frac{1}{s - \zeta}\right\} = e^{\zeta t}.$$

Therefore, equation (4.2) becomes

$$f(t) = \frac{1}{2\pi i} \lim_{\beta \to \infty} \int_{\gamma - \beta i}^{\gamma + \beta i} F(\zeta) e^{\zeta t} d\zeta \tag{4.3}$$

□

Theorem 4.3 (Sufficient Conditions for Inversion)**.** *Let F be a function of complex variable s that is analytic for all $s = x + iy$ in the half-plane $\alpha < s$, and $F(s)$ is real-valued when $x > \alpha$. Further, let $k > 1, M$, and r be constants such that $|F(s)s^k| < M$ for $|s| > r$ in the half-plane $x > \alpha$. Then the inversion integral of F along any line $x = \gamma$ defined by*

$$\frac{1}{2\pi i} \lim_{\beta \to \infty} \int_{\gamma - \beta i}^{\gamma + \beta i} e^{st} F(s) ds, \tag{4.4}$$

converges to a real-valued function t.

Proof. Since $F(s)$ is analytic in the half-plane $\alpha < s$, we know that $e^{st}F(s)$ is a continuous function of y and t, where $z = \gamma + iy$ and $\gamma > \alpha$. Also since $F(s)$ is of exponential order k, it follows that there exists a positive real number y_0 such that

$$|F(\gamma + iy)| < \frac{M}{(\gamma^2 + y^2)^{\frac{k}{2}}} \leq \frac{M}{|y|^k} \qquad \text{when } |y| > y_0. \tag{4.5}$$

Further, since $F(s)$ is analytic and $F(x)$ is real in the half-plane $x > \alpha$, we have $F(\overline{s}) = \overline{F(s)}$ by the Schawrz reflection principle, and therefore the inversion integral takes the real form:

$$\frac{e^{\gamma t}}{\pi} \int_0^\infty \mathrm{Re}[e^{iyt} F(\gamma + iy)] dy$$

$$= \frac{e^{\gamma t}}{\pi} \int_0^{y_0} \mathrm{Re}[e^{iyt} F(\gamma + iy)] dy + \frac{e^{\gamma t}}{\pi} \int_{y_0}^\infty \mathrm{Re}[e^{iyt} F(\gamma + iy)] dy$$

Now by the condition (2.6), we obtain

$$|\mathrm{Re}[e^{iyt} F(\gamma + iy)]| < 2M|y^k| \qquad \text{for } |y| > y_0.$$

Hence the improper integral above converges uniformly with respect to t, and since $\mathrm{Re}[e^{iyt} F(\gamma + iy)]$ is continuous for all t and y, we can conclude that both integrals converge and represent real-valued functions of t. □

Another important result for inverting a Laplace transform is given below.

Theorem 4.4. *Let F be a function for which the inversion integral along a line $x = \gamma$ represents the inverse function f, and let F be analytic except for isolated singularities s_n $(n = 1, ...)$ in the half-plane $x < \gamma$. Then the series of residues of $e^{st}F(s)$ at $s = s_n$ converges to f for each positive t, provided a sequence C_n of contours can be found that satisfies the following properties:*

1. *C_n consists of the straight line $x = \gamma$ from $\gamma - \beta_n i$ to $\gamma + \beta_n i$ and some curve Γ_n beginning at $\gamma + \beta_n i$, ending at $\gamma - \beta_n i$, and lying in $x \leq \gamma$.*
2. *C_n encloses $s_1, s_2, ..., s_n$.*
3. $\lim_{n \to \infty} \beta_n = \infty$

(4) $\lim_{n\to\infty} \int_{\Gamma_n} e^{st} F(s) ds = 0$

Proof: *Refer to* [5]

EXAMPLE: NON-HOMOGENEOUS WAVE EQUATION ON A BOUNDED DOMAIN

To conclude this paper, we will consider a non-homogeneous PDE and use the method of Laplace transforms to obtain a solution. In the process, we will be forced to use the inversion formula to obtain the final solution.

Example 4.5. A taut string of length L initially at rest has both its ends fixed. If a magnetic field of strength $F_0 \sin(\omega t)$ (where F_0 and ω are positive constants and ω is not an integral multiple of $\frac{\pi c}{L}$) is activated at the initial moment, the mathematical model for the displacement u(x,t) of point in the string is as follows:

$$\frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2} + F_0 \sin \omega t, \quad 0 < x < L, \ t > 0; \tag{4.6a}$$

$$u(0,t) = 0, \quad t > 0; \tag{4.6b}$$

$$u(L,t) = 0, \quad t > 0; \tag{4.6c}$$

$$u(x,0) = 0, \quad 0 < x < L \tag{4.6d}$$

$$\frac{\partial u}{\partial t} = 0, \quad 0 < x < L \tag{4.6e}$$

Solution. To begin, we take the Laplace transform of equation (4.6) to obtain:

$$s^2 U(s) - su(0) - u_t(0) = c^2 \frac{\partial^2}{\partial x^2} \mathcal{L}\{u\} + F_0 \frac{\omega}{s^2 + \omega^2}$$

Applying the boundary conditions (4.9) and (4.10), we get:

$$s^2 U(x,s) = c^2 \frac{\partial^2 U(x,s)}{\partial x^2} + \frac{F_0 \omega}{s^2 + \omega^2}, \quad 0 < x < L \tag{4.7}$$

or

$$\frac{\partial^2 U}{\partial x^2} - \frac{s^2}{c^2} U = -\frac{F_0 \omega}{c^2 (s^2 + \omega^2)}, \quad 0 < x < L \tag{4.8a}$$

This is now an ordinary differential equation that is subject to the transformed condition:

$$U(0,s) = 0 \tag{4.8b}$$

$$U(L,s) = 0 \tag{4.8c}$$

The homogeneous solution to this equation is

$$U(x,s) = A \cosh\left(\frac{sx}{c}\right) + B \sinh\left(\frac{sx}{c}\right)$$

We find a particular solution by assuming that $U_x x = 0$, this will give

$$U(x,s) = \frac{F_0 \omega}{s^2 (s^2 + \omega^2)}$$

Therefore, the general solution will be:

$$U(x,s) = A\cosh\left(\frac{sx}{c}\right) + B\sinh\left(\frac{sx}{c}\right) + \frac{F_0\omega}{s^2(s^2+\omega^2)} \tag{4.9}$$

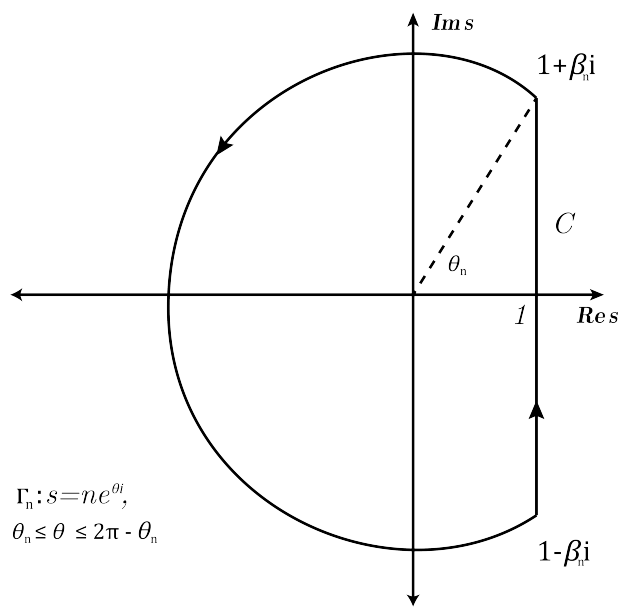
Applying the boundary condition (4.8b) and (4.8c), we obtain:

$$0 = A + \frac{F_0\omega}{s^2(s^2+\omega^2)}, \qquad 0 = A\cosh\left(\frac{sL}{c}\right) + B\sinh\left(\frac{sL}{c}\right) + \frac{F_0\omega}{s^2(s^2+\omega^2)}$$

Hence,

$$\begin{aligned} U(x,s) = &-\frac{F_0\omega}{s^2(s^2+\omega^2)}\cosh\left(\frac{sx}{c}\right) + \frac{F_0\omega}{s^2(s^2+\omega^2)\sinh(sL/c)}\times \\ &\left(-1+\cosh\left(\frac{sx}{c}\right)\right)\sinh(sx/c) + \frac{F_0\omega}{s^2(s^2+\omega^2)} \end{aligned} \tag{4.10}$$

To invert this, we find the residues of $e^{st}U(x,s)$ at its singularities using the contour below.



By applying Theorem 3.1, it is easy to see that this contour will satisfy conditions 1-4 of Theorem 4.4, hence $f(t)$ can be found by taking the sum of residues of $e^s tF(s)$ at its singularities.

To determine the type of singularity at $s = 0$, we expand $U(x,s)$ in a Laurent expansion around $s = 0$. This is given by

$$U(x,s) = \frac{F_0\omega}{s^2(s^2+\omega^2)}\Bigg[1 - \left(1 + \frac{s^2x^2}{2c^2} + \frac{s^4x^4}{24c^4} + ...\right)$$
$$+\left(\frac{s^2L^2}{2c^2} + \frac{s^4L^4}{24c^4} + ...\right)\left(\frac{\frac{sx}{c} + \frac{s^3x^3}{6c^3} + ...}{\frac{sL}{c} + \frac{s^3L^3}{6c^3} + ...}\right)\Bigg]$$

$$= \frac{F_0\omega}{s^2(s^2+\omega^2)}\Bigg[-\left(\frac{s^2x^2}{2c^2} + \frac{s^4x^4}{24c^4} + ...\right)$$
$$+\left(\frac{s^2L^2}{2c^2} + \frac{s^4L^4}{24c^4} + ...\right)\left(\frac{x}{L} + \frac{s^2x(x-L)}{6Lc^2} + ...\right)\Bigg]$$

$$= \frac{F_0\omega x(L-x)}{2c^2(s^2+\omega^2)} + ...$$

Therefore $U(x,s)$ will have a removable singularity at $s = 0$ and this will lead immediately to the conclusion that $e^{st}U(x,s)$ also has a removable singularity at $s = 0$. Next we consider the residue at $s = i\omega$. It is easy to see that this will be a simple pole, and if we apply the residue formula, we find:

$$Res(e^{st}U(x,s), i\omega) = \lim_{s\to i\omega}(s - i\omega)\cdot e^{st}\cdot U(x,s)$$

$$= \lim_{s\to i\omega}(s - i\omega)\cdot e^{st}\cdot\Bigg(-\frac{F_0\omega}{s^2(s^2+\omega^2)}\cosh\left(\frac{sx}{c}\right) + \frac{F_0\omega}{s^2(s^2+\omega^2)\sinh(sL/c)}\times$$
$$\left(-1 + \cosh\left(\frac{sx}{c}\right)\right)\sinh(sx/c) + \frac{F_0\omega}{s^2(s^2+\omega^2)}\Bigg)$$

$$= \lim_{s\to i\omega}\cdot e^{st}\cdot\Bigg(-\frac{F_0\omega}{s^2(s+i\omega)}\cosh\left(\frac{sx}{c}\right) + \frac{F_0\omega}{s^2(s+i\omega)\sinh(sL/c)}\times$$
$$\left(-1 + \cosh\left(\frac{sx}{c}\right)\right)\sinh(sx/c) + \frac{F_0\omega}{s^2(s+i\omega)}\Bigg)$$

$$= -e^{i\omega t}\cdot\Bigg(-\frac{F_0\omega}{2i\omega^3}\cosh\left(\frac{i\omega x}{c}\right) + \frac{F_0\omega}{2i\omega^3\sinh(i\omega L/c)}\left(-1+\cosh\left(\frac{i\omega x}{c}\right)\right)\sinh(sx/c) + \frac{F_0\omega}{2i\omega^3}\Bigg)$$

$$= \frac{F_0e^{i\omega t}}{2i\omega^2}\cdot\left(\cosh(i\omega x/c) + \frac{\sinh(sx/c)}{\sinh(i\omega L/c)} - \frac{\cosh(i\omega x/c)\sinh(sx/c)}{\sinh(i\omega L/c)} - 1\right)$$

By applying Euler's identity and simplifying, we obtain:

$$\frac{F_0 e^{i\omega t}}{2\omega^2 \sin(\omega L/c)} \cdot \left(\sin(\omega L/c)\Big[\cos(\omega x/c) - 1\Big] - \sin(\omega x/c)\Big[\cos(\omega L/c) - 1\Big] \right)$$

and using the difference formula for sine, we get

$$\frac{F_0 e^{i\omega t}}{2\omega^2 \sin(\omega L/c)} \cdot \left(\sin(\omega x/c) + \sin(\omega L/c) + \sin(\omega(L-x)/c) \right) \tag{4.11}$$

In a similar fashion, we find

$$Res(e^{st}U(x,s), -i\omega) = \lim_{s\to -i\omega} (s+i\omega)\cdot e^{st}\cdot U(x,s)$$

$$= \lim_{s\to -i\omega} (s+i\omega)\cdot e^{st}\cdot \Bigg(-\frac{F_0\omega}{s^2(s^2+\omega^2)}\cosh\left(\frac{sx}{c}\right) + \frac{F_0\omega}{s^2(s^2+\omega^2)\sinh(sL/c)}\times \left(-1+\cosh\left(\frac{sx}{c}\right)\right)\sinh(sx/c) + \frac{F_0\omega}{s^2(s^2+\omega^2)}\Bigg)$$

$$= \lim_{s\to i\omega} \cdot e^{st}\cdot \Bigg(-\frac{F_0\omega}{s^2(s-i\omega)}\cosh\left(\frac{sx}{c}\right) + \frac{F_0\omega}{s^2(s-i\omega)\sinh(sL/c)}\times \left(-1+\cosh\left(\frac{sx}{c}\right)\right)\sinh(sx/c) + \frac{F_0\omega}{s^2(s-i\omega)}\Bigg)$$

$$= e^{-i\omega t}\cdot\Bigg(-\frac{F_0\omega}{-2i\omega^3}\cosh\left(\frac{-i\omega x}{c}\right) + \frac{F_0\omega}{-2i\omega^3\sinh(i\omega L/c)}\left(-1+\cosh\left(\frac{i\omega x}{c}\right)\right)\sinh(-i\omega x/c) + \frac{F_0\omega}{2i\omega^3}\Bigg)$$

$$= \frac{F_0 e^{-i\omega t}}{2i\omega^2}\cdot\left(\cosh(i\omega x/c) + \frac{\sinh(sx/c)}{\sinh(i\omega L/c)} - \frac{\cosh(i\omega x/c)\sinh(sx/c)}{\sinh(i\omega L/c)} - 1\right)$$

Once again, using Euler's identity and simplifying, we obtain:

$$\frac{F_0 e^{-i\omega t}}{2\omega^2 \sin(\omega L/c)} \cdot \left(\sin(\omega L/c)\Big[\cos(\omega x/c) - 1\Big] - \sin(\omega x/c)\Big[\cos(\omega L/c) - 1\Big] \right)$$

and using the difference formula for sine, we get

$$\frac{F_0 e^{-i\omega t}}{2\omega^2 \sin(\omega L/c)} \cdot \left(\sin(\omega x/c) + \sin(\omega L/c) + \sin(\omega(L-x)/c) \right) \tag{4.12}$$

Combining (4.11), (4.12), and the result for $Res(e^{st}U(x,s), 0)$, we get:

$$
\begin{aligned}
&Res(e^{st}U(x,s),0) + Res(e^{st}U(x,s),i\omega) + Res(e^{st}U(x,s),-i\omega) \\
&= \frac{F_0}{\omega^2 \sin(\omega L/c)} \cdot \Bigg(\sin(\omega x/c) + \sin(\omega L/c) + \sin(\omega(L-x)/c) \Bigg) \sin(\omega t)
\end{aligned} \tag{4.13}
$$

Finally, we must compute the residues of $e^{st}U(s,x)$ at $i(2n-1)\pi c/L$ where $n \in \mathbb{Z}$. This will be

$$
Res(e^{st}U(x,s), i(2n-1)\pi c/L) = \lim_{s \to i(2n-1)\pi c/L} (s - i(2n-1)\pi c/L) \cdot e^{st} \cdot U(x,s)
$$

$$
= \lim_{s \to i(2n-1)\pi c/L} (s - i(2n-1)\pi c/L) \cdot e^{st} \cdot \Bigg[- \frac{F_0 \omega}{s^2(s^2+\omega^2)} \cosh\left(\frac{sx}{c}\right)
$$

$$
+ \frac{F_0 \omega}{s^2(s^2+\omega^2)\sinh(sL/c)} \Bigg(-1 + \cosh\left(\frac{sx}{c}\right)\Bigg) \sinh(sx/c) + \frac{F_0 \omega}{s^2(s^2+\omega^2)} \Bigg]
$$

$$
= \lim_{s \to i(2n-1)\pi c/L} (s - i(2n-1)\pi c/L) \cdot e^{st} \cdot \Bigg[\frac{F_0 \omega \sinh(sx/c)}{s^2(s^2+\omega^2)\sinh(sL/c)} \Bigg(-1 + \cosh\left(\frac{sx}{c}\right)\Bigg)\Bigg]
$$

Applying L'Hospital's rule to the limit above, we get

$$
\frac{4F_0 \omega L^3}{2i(c\pi^2)(2n-1)^2[\omega^2 L^2 - (2n-1)^2\pi^2 k]} \sin\left(\frac{(2n-1)\pi c x)}{L}\right) e^{\left(\frac{i(2n-1)\pi c t}{L}\right)} \tag{4.14}
$$

Next notice that:

$$
\sum_{n \in \mathbb{Z}} \frac{4F_0 \omega L^3}{2i(c\pi^2)(2n-1)^2[\omega^2 L^2 - (2n-1)^2\pi^2 k]} \sin\left(\frac{(2n-1)\pi c x)}{L}\right) e^{\left(\frac{i(2n-1)\pi c t}{L}\right)} \tag{4.15}
$$

$$
= \frac{4F_0 \omega L^3}{(c\pi^2)} \sum_{n=1}^{\infty} \frac{1}{(2n-1)^2[\omega^2 L^2 - (2n-1)^2\pi^2 k]} \sin\left(\frac{(2n-1)\pi c x)}{L}\right) \sin\left(\frac{i(2n-1)\pi c t}{L}\right)
$$

If we combine this with the result (4.13), we obtain the final solution of:

$$
\frac{F_0}{\omega^2 \sin(\omega L/c)} \cdot \Bigg(\sin(\omega x/c) + \sin(\omega L/c) + \sin(\omega(L-x)/c) \Bigg) \sin(\omega t) \tag{4.16}
$$

$$
+ \frac{4F_0 \omega L^3}{(c\pi^2)} \sum_{n=1}^{\infty} \frac{1}{(2n-1)^2[\omega^2 L^2 - (2n-1)^2\pi^2 k]} \sin\left(\frac{(2n-1)\pi c x)}{L}\right) \sin\left(\frac{i(2n-1)\pi c t}{L}\right)
$$

Conclusion

I enjoyed this project because I learned a great deal about complex analysis and its role in finding the inverse Laplace transform of a function. I also found it interesting how the Laplace transform is used to solve an initial boundary value problem. I found the problem that was assigned to be very challenging in the terms of the computation of residues. There have been a few steps that were shortened in this paper in order to finish in a reasonable length. I spent many hours checking and rechecking work by hand, and I hope that in writing this document I have not made any mistakes. With that being said, I have checked my work against the answer provided in [5] and it does indeed agree with the provided solution. I hope that future students will also get the chance to work on such interesting problems.

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